DR. NII AMARQUAYE COMMEY

GEOSPATIAL & ENVIRONMENTAL SCIENTIST | REMOTE SENSING, WATER SYSTEMS & ENVIRONMENTAL MODELLING

Email | LinkedIn | Portfolio | Google Scholar

PROFESSIONAL PROFILE

Geospatial and environmental scientist with a PhD in Engineering (University of Yamanashi) and more than a decade of experience across academia, environmental consulting and international research collaboration. Uses Earth observation, GIS and environmental modelling to turn satellite, field and hydrological data into evidence for water, coastal and land-management decisions. Work includes multi-decadal land-use change analysis of Ramsar-listed catchments in Ghana, watershed pollution-load research, and cloud-based satellite analysis in Google Earth Engine. Currently coordinating an international Japan–sub-Saharan Africa education and research programme, and seeking roles in spatial data analysis, climate resilience, water systems and sustainable land management.

TECHNICAL SKILLS

GIS & Spatial Analysis: ArcGIS Pro, QGIS, ArcPy, raster and vector processing, catchment delineation, multi-temporal analysis, change detection, cartography.

Earth Observation: Google Earth Engine, optical and radar (SAR) imagery, satellite time-series analysis, land-use and land-cover classification, environmental monitoring.

Environmental & Hydrological Modelling: Rainfall–runoff analysis, non-point-source pollution load estimation (QNSPECT), watershed assessment, wetland monitoring, flood-related research.

Data & Reproducible Research: Python, R, multi-source data integration, statistical trend analysis, documented and reproducible workflows.

Assessment & Coordination: Environmental Impact Assessments and Management Plans, field-survey coordination, technical reporting, stakeholder communication, international programme coordination.

PROFESSIONAL EXPERIENCE

International Programmes Coordinator; University of Yamanashi, Japan | Apr. 2026 – Present

- Coordinate CoDeWS, an international education and research programme linking the University of Yamanashi, Kitasato University and the University of Ghana, focused on sustainable well-being in sub-Saharan Africa.
- Support cross-institutional communication, documentation and delivery, including the September 2026 programme kickoff and the 2026 student exchange in Japan.

Research Assistant Professor; University of Yamanashi, Japan | Apr. 2025 – Mar. 2026

- Conducted research on watershed pollution-load estimation, integrating stable-isotope ($\delta^{15}N$, $\delta^{18}O$), hydrological, land-use and demographic data to identify pollutant sources and watershed-scale processes.
- Worked toward reproducible, open-source Python workflows for pollution-load estimation.
- Mentored graduate researchers and led manuscript drafting.

Satellite Data Analysis Engineer; Solafune, Japan | Nov. 2024 – Mar. 2025

- Developed Google Earth Engine workflows for satellite time-series and change-detection analysis in environmental, agricultural and climate monitoring projects.
- Contributed to machine-learning-based land-use and change-detection pipelines and translated satellite outputs into decision-ready datasets for client applications.

Research & Teaching Assistant; University of Yamanashi, Japan | Sep. 2024 – Mar. 2025

- Supported graduate research in environmental data analysis and geospatial methods, including workflow refinement, data sourcing, and manuscript and presentation preparation.

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PROFESSIONAL EXPERIENCE

Environmental Consultant & Marine Biologist; ESL Consulting, Ghana | Sep. 2015 – Oct.2021

- Contributed to Environmental Impact Assessments and Environmental Management Plans for port, LNG and offshore energy developments.
- Coordinated vessel-based field surveys (benthic ecology, fish, sediment, water quality) with field teams of three to eight people.
- Prepared technical reports and mitigation recommendations for regulators and project proponents.

Teaching Assistant; University of Ghana, Ghana | Sep. 2014 – Aug. 2015

- Supported undergraduate teaching and practicals in marine and environmental science, including cohorts of more than 200 students, and co-supervised three undergraduate research projects.

PROFESSIONAL EXPERIENCE

Ph.D. in Engineering (Environmental and Social System Science), University of Yamanashi, Japan | 2025 Thesis: Assessing the Effects of Catchment Changes on the Sakumo and Songor Ramsar Sites in Ghana Using Geospatial Techniques

M.Phil. in Marine Science, University of Ghana, Ghana | 2017 Thesis: Macrobenthic Infaunal Assemblage Structure Analysis

B.Sc. in Oceanography, University of Ghana, Ghana | 2014 Project: Investigation of Upwelling Dynamics Using Satellite Data

SELECTED PUBLICATIONS

1. **Commey, N.A.**, Magome, J., Ishidaira, H., & Souma, K. (2025). Assessment of Rainfall-Runoff and Non-Point Source Pollution Loads in the Sakumo Catchment using GIS and QNSPECT. *Journal of JSCE*.
2. Wei, Y., Li, H., Zhou, Y., **Commey, N.A.**, Ishidaira, H., & Yang, D. (2025). Flood vulnerability assessment of urban agglomeration under the background of urban expansion. *Journal of Hydrology*.
3. Li, H., Wei, Y., Ishidaira, H., **Commey, N.A.**, & Yang, D. (2024). Integrated urban and riverine flood risk management in the Fujiang River Basin-Mianyang city. *Journal of Hydrology*.
4. **Commey, N.A.**, Magome, J., Ishidaira, H., & Souma, K. (2023). Catchment-Scale Land Use and Land Cover Change Analysis in Two Coastal Ramsar Sites in Ghana, Using Remote Sensing. *Water*.

Full publication record on Google Scholar.

AWARDS & PROFESSIONAL ENGAGEMENT

- MEXT Graduate Scholarship, Japan (2021)
- ICRE Award, Online Salon (2023)
- Training: ESA and NASA ARSET Earth Observation and remote sensing programmes; Esri GIS and geospatial training; Environmental and Social Impact Assessment

This condensed CV highlights key qualifications. The full CV available upon request.